

World Model for Indian Cities

Problem Statement

We started with a problem, how we can find the bottleneck of Indian cities like Delhi and why these points are considered as bottle neck. How we can improve these bottle necks to improve overall flow of traffic in cities. Finding a bottleneck can be challenging since we cannot visit all roads/junctions/lanes individually, and find out which road is acting as a bottle neck for that area. Google maps provides us the traffic status in the form of colours like red, orange, yellow, green. But most of the time, the user does not know the cause why they have traffic on that road. Possible causes can be traffic signal, accidents, construction work, metro station pick up/ drop point, shared mobility points near subway, or it can be illegal encroachments on roadside, maybe we have some fuel station where the vehicles have a long queue which comes on road, reduces the effective area for traffic flow. One reason is religious places where we can see traffic occasionally, it can be due to parking of vehicles on roads near places. Also at tourist places we can see traffic, possibly due to local markets, autos, etc which causes problems to people who travel through that area. But we know that these places like tourist spots, religious places, the subway, and shared mobility points are important things for any city. So to improve traffic at these places, we just need a proper management of traffic. We have another problem also, that what traffic colour we see in Google Maps is representing the correct traffic status present at that junction at that time. Suppose we see a 100 m red in Google Maps, so this 100 m is actually present in the road or it is more/less than 100 m what we have on the road. We know that in India many junctions have fixed traffic light time, and suppose if we have different timing acc to peak hours, we do not know that in actually those signal timings are the optimized one that we can have for that junction or we can improve that signal timings so that we have a good traffic flow in junction in compare what we have now?. This traffic signal timing is one of the major problems when we talk about traffic in an area. Since if these timings are not optimized, like we have two lanes with equal green light signal to allow vehicles to cross, but if one of lanes have only 2-10 vehicles(less traffic) then that green timing is not efficient for that road, instead it causes more vehicles to stop at opposite lane, which result in congestion at another road, increasing waiting time, fuel wastage, higher emissions, also sometimes leads to driver frustration which leads driver to cross that signal illegally. So we have understood some of the problems that we can have in Delhi, also similar to Delhi we can have these problems in many other states also

Work that we did

To approach these problems, we use Google Maps real traffic status, take screenshots to find some meaningful way how we can solve the above stated problems. I start with a basic task to find around 10 junctions and find bottlenecks for those junctions at a particular time interval. We mainly do this for peak hour. I do this analysis on south-west Delhi covering the area of Aiims, Bhikaji Cama place, Hauz Khas, Moolchand, Pansheel park, Chirag Delhi, Greater Kalish, Nehru Place, Okhla NSIC, Lajpat Nagar. I used to take screenshots of these areas for around 2-5 days and take those roads which have consistent red through intervals and those roads which suddenly turned red in peak hours but green/yellow at non peak hours. These analyses give me an overall idea of the possible bottle neck for these areas. Then I physically visit these

areas to actually find why these roads are the breaking point(bottleneck) for that junction. I tried to frame a traffic cause list for Delhi. I find that in Delhi major causes of traffic are, in some areas roads are merging/diverging which causes traffic, like in Bhikaji Cama Place- VMMC road, I find that we have 6-7 lanes which are merging into 3 lanes which causes major traffic on the road. So this is due to the geometry of roads which is a major cause of traffic at many places in Delhi. In addition to this, other causes are illegal encroachments inside of roads, poor management of traffic around markets and tourist/religious places. In some areas traffic signal timing is either fixed over day or if it is changing over time, then in most cases it is not optimized, it can be better by doing more analysis of that area. In many junctions, I find that a free left turn at a traffic light can result in overall throughput coming from the straight road to that left turn road. Next task we try to find queue length at different traffic light junctions, since physically visiting all area and measure length is not the way we can do this task, so I try to use Google Maps to do this task, but we are not sure that what length we got through map is the exact that present in real scenario at that time. So we try to generalize that the red colour length that we got from the map is the real or it is the upper or lower bound of the real one?. So for this task we physically visit Hauz khas -IIT D junctions by taking readings of queue length from real and Google Maps simultaneously. So we got that Google Maps slightly overestimates the queue length compared to field observations, but not much larger, which can say that it is giving us the weak upper bound of the real queue length. In addition to the upper bound, if traffic is developed instantaneously, then Google Maps will update in a span of 2-5 minutes. Since we get that the queue length is not the same as the real one, we analyze the ratio of queue length of all lanes present in one junction. Our finding is that the average difference between queue length measured from Google Maps and real one over a time interval is approximately 4.7%. So we can use the ratio of queue length calculated from Google Maps for our future work. After coming to the conclusion of the queue length ratio, we decide to allocate green signal timing from the ratio. So we take a fixed cycle length of 260 seconds, in which 20 seconds is for yellow time if we have four lanes in a junction. We try to allocate green signal time for each lane using the formula:

$$\text{Green_time} = \text{Queue_length_ratio_of_lane} * 240$$

subject to constraint that the minimum green time for each lane is 10 sec. So by doing this we got different green signal timing for a different time stamp. We measure the queue ratio in an interval of 5 minutes. So by having queue length ratio at each time stamp as input, we calculate green signal timing for each lane at each time stamp. Then we use both green signal timing, the one we got from ratio data and the other one is the real one that is now functioning at the junction at that time interval. We simulate the queue length with both green signal timing for each cycle, by taking a constant discharge rate of 2.5m/sec. By doing this, we have the remaining queue, maximum queue, queue reduction for signal time (adaptive time, we got this from the ratio and another is fixed time, which is the real one at the junction). By plotting these comparisons, we found that our adaptive timing for Hauz Khas junction is better than present timing, given at the same queue length over the same time interval.

Also we did one more thing from this queue ratio data, we compare the opposite roads also at Hauz khas junction. Two Pairs were formed for Hauz khas junction, then in each pair we take

the ratio of queue length of opposite roads and ratio of adaptive green signal in the same order. So we have two ratios of each pair, one is the queue length ratio and green signal ratio of opposite roads. Then we plot a graph of the queue length ratio and the green signal ratio, we use both the green signal ratio (adaptive and fixed) and compare which suits well with the queue length ratio. We found that our adaptive timing performs better than the existing fixed timing under the assumptions of our simulation.

What We Got

So by doing all the above analysis over Huaz Khas Junction, we got to know that green signal allocation at this junction for each lane can be optimized further for reducing congestion. In the future we want to check if the result that we got from our analysis is consistent with the real scenario or not. According to our analysis, if our result is consistent with the real scenario, then at Hauz Khas we can have 6.27% reduction in the remaining queue and approx 24.6% reduction in the maximum queue.

Average Remaining Queue (Adaptive): 613.33 m

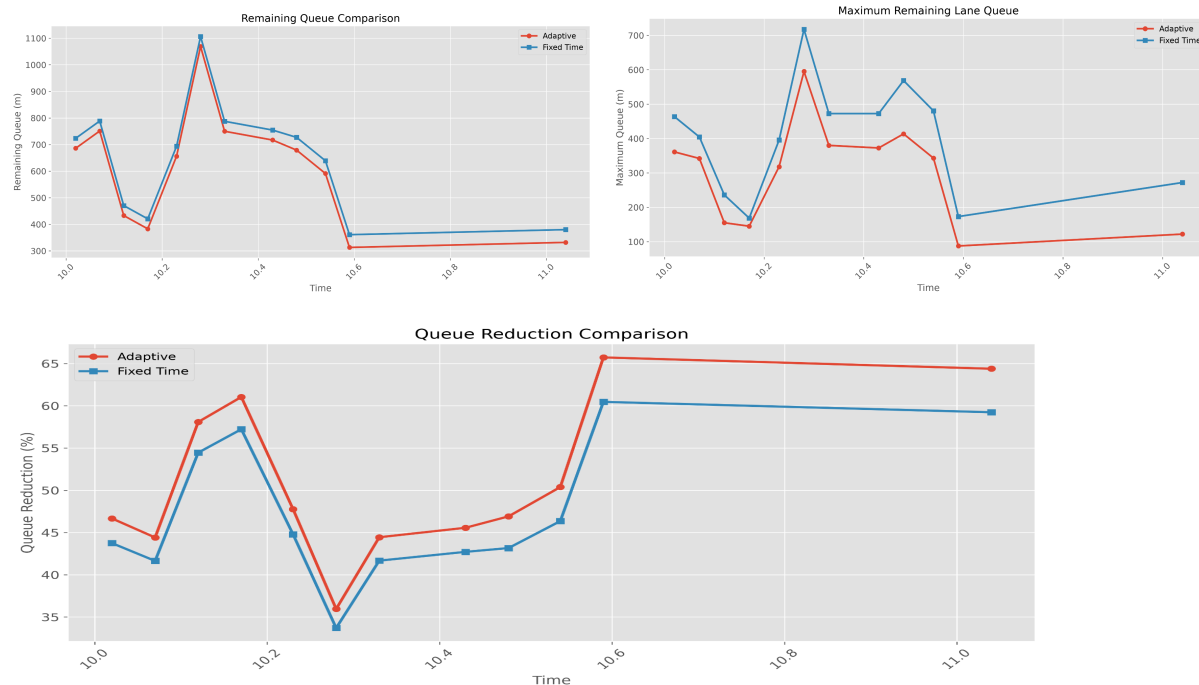
Average Remaining Queue (Fixed): 654.33 m

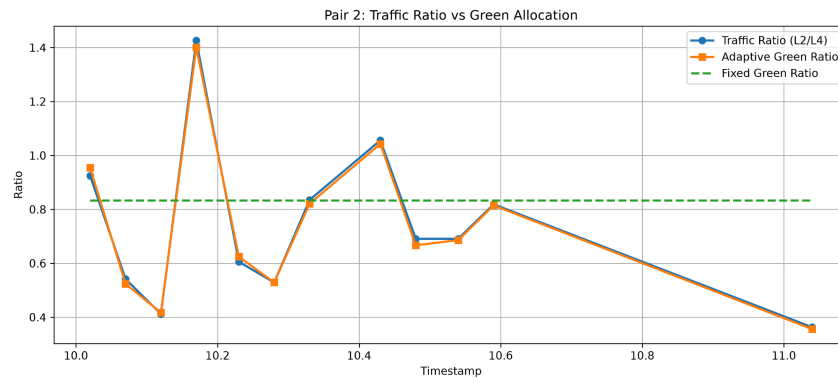
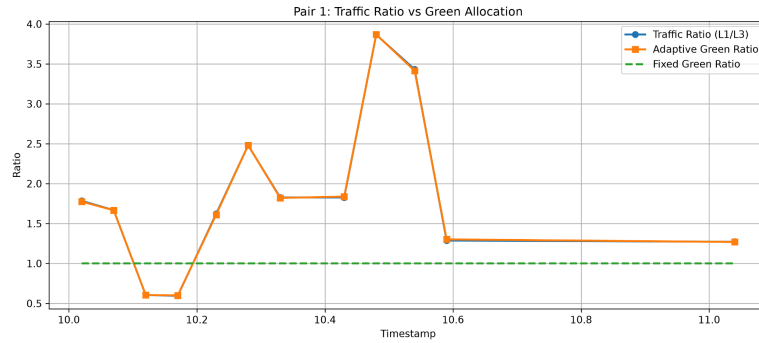
Average Maximum Queue (Adaptive): 302.88 m

Average Maximum Queue (Fixed): 402.04 m

Remaining Queue Improvement : 6.27%

Maximum Queue Improvement : 24.67%





Future Goals

If it is consistent, then we can do this analysis for other junctions in Delhi also. We can scale this analysis over many junctions, by doing these tasks to be more automated, like for now we do this 1-2 junction only, so we can measure queue length manually from Google Maps screenshot. We are trying to automate this task of calculating queue length from Google Maps captures. This automation required more things, not just masking of image will work, since in map capture we have extra things also so we need to take care of them also. We want to automate this task but have to first check how much accurate data we got from this automation. Also for Simulation, for now we just wrote a python script to do this, in future to better visualize this we can use more advanced simulation tools like SUMO.